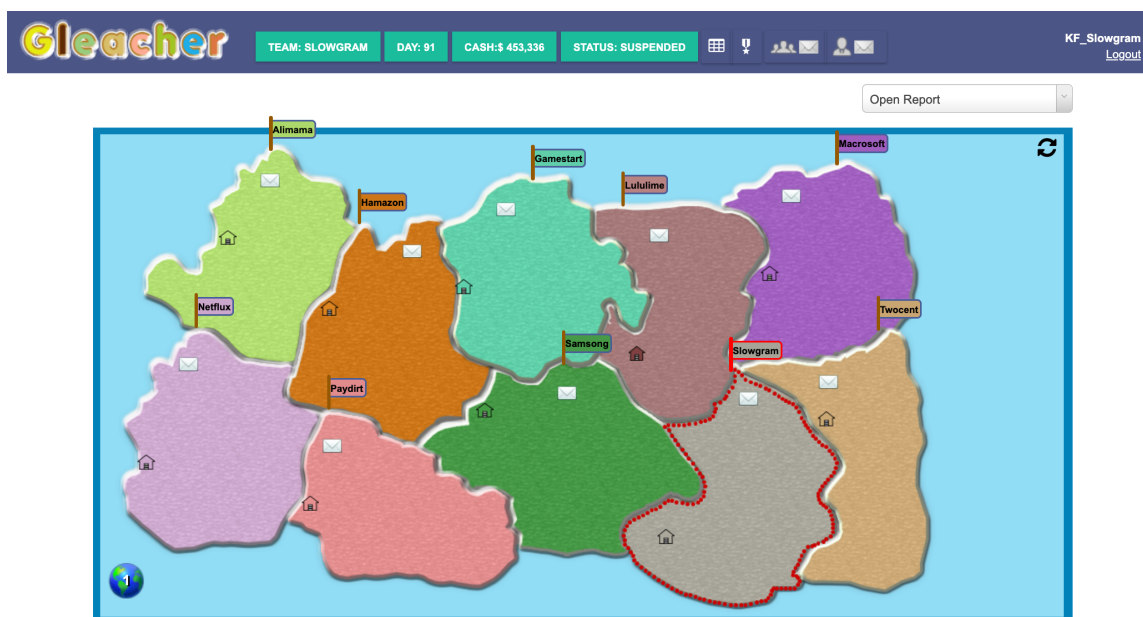


Trading Game Case Brief

Overview

Welcome to the Monopoly Game. Your team will be managing a firm that manufactures and sells two wearable medical monitors as a monopolist in its own region. The first product, named “Hormone”, tracks the concentration of several hormones. It is a generic product that any firm can make. The second product is a proprietary specialty good.

After logging in to the game, you will see:



The top row shows your team name, the game day, the current cash balance, the game status, your username, and the logout button. The game status may be suspended, running, or done.

Your home region is outlined in red dots on the map and contains a distribution center (DC) icon that opens a panel to manage inventory and the pricing of your hormone and specialty products.

There are four controls above the map:

- The “History” button opens a table showing all the actions taken by your team.
- The “Scoreboard” button shows all the teams ranked by cash position.
- The “Chat” buttons allow open chat windows for you to communicate with all teams or admin.
- The “Open Report” dropdown opens a list of useful reports (see the *Reports* section below).

Background

The firm manufactures and sells two wearable medical monitors. It was founded 91 days ago by an engineer who designed the products. He contributed his life savings of \$3,000,000 to the firm and used the funds to purchase a factory for production and a distribution center to store and ship the products. He was struggling with business management, when you and your team came along and paid him \$1,500,000 for 50% of the firm. Your team will now take over managing the business.

Products are manufactured in batches of 100. The materials cost for each product is \$100. Production of a batch begins by incurring a cost of $\$100 * 100 \text{ units} = \$10,000$. The batch takes 2.5 days to manufacture, and the factory can only make one batch of a single product at a time. Once the batch is completed, another batch can begin production while the finished batch goes into “transit” for 24 hours as it is sent from your factory to your distribution center (DC). Shipping from your factory to your DC is free. When shipping between regions, it takes 3 days and costs \$40/unit to ship by mail, and it takes 21 days and costs \$10,000 per container. Mail is shipped in sets of 10 units. Each container holds 1,000 units. Once the batch arrives at a DC, it goes into on-hand inventory. All sales are made from on-hand inventory at the DC. The factory and DC operate 24 hours a day.

The market size for the hormone product is 300,000. The market size for the specialty product in its home region is 140,000, but market size varies across other regions. Retail customers for each product arrive at the DC daily at a constant rate of $0.0001 * \text{the market size}$. The willingness to pay (WTP) of individual customers is uniformly distributed along a range of \$0 to maximum WTP. The maximum WTP for a product is the same across all regions. You set a separate retail price for each product. If a retail customer for a particular product arrives and that product’s price is higher than what the customer is willing to pay, the customer is lost forever. If a retail customer for a particular product arrives and there is no on-hand inventory to sell, the customer is also lost forever.

There is a shipping agreement for each product that is manufactured at the factory and sent to your DC. You control production by setting a reorder point for each incoming shipping agreement. The DC asks the factory to make a new batch when the sum of in-process, in-transit, and on-hand inventory is equal to or less than your reorder point. You can turn off production by setting the reorder point to -1. If production of both products is requested simultaneously, then the facility alternates producing a batch of one product then a batch of the other product.

Products can also be sold from one team to another using shipping agreements. The selling team creates a shipping agreement and sends the proposal to the buying team. Teams can propose and counter propose revisions to the shipping agreement prior to accepting the agreement. See the Shipping Agreement section below for an overview of possible terms. Once the buying team accepts the latest proposal sent by the selling team, the agreement becomes active and can no longer be revised, but it can be terminated by either party.

- Once a shipping agreement between two teams is active, the terms cannot be changed.
- The buying team orders inventory by setting a reorder point that triggers the shipment of inventory from the selling team’s DC.
- The buying team can change an agreement’s reorder point at any time.
- The selling team’s DC does not ship until it has enough on-hand inventory to meet the order quantity specified in the shipping agreement terms.

- Each shipping agreement in the Outgoing Shipping Agreements table has a priority setting that determines which agreement gets processed first if there not enough inventory to satisfy two or more simultaneous orders.
- The agreement with the highest priority is shipped first. Higher values = higher priorities (e.g., 2 comes before 1).
- If shipping agreements with identical priorities are triggered at the same time, the agreement that has waited the longest since it was either activated or triggered previously is shipped.
- The selling team can change an agreement's priority at any time.
- Shipping charges are incurred by the selling team.

At the end of the game the monitor technology will be replaced by a new generation of technology. As result the company shuts down and all inventory becomes obsolete. Your objective is to maximize the firm's cash balance at the end of the game, practice trading with other teams, and optimize wholesale prices and supply chains.

Distribution Center Controls

Click the DC icon  in your region to open a panel to manage the inventory for all of your products and trade with other teams. The panel does not auto-refresh; it can be manually refreshed using the refresh icon at the top right. You will see the following icons:

-  opens a panel to show more details
-  refreshes the panel
-  next to a column label reveals additional columns of data
-  lets you to change a parameter
-  opens a daily plot of some metric
-  downloads data from a report
-  lets you to clone an outgoing shipping agreement

Change retail prices by clicking  next to retail price in the Inventory table of the DC.

Change reorder points for an incoming shipping agreement by clicking  next to the agreement in the Incoming Shipping Agreements table.

Change the priority of an outgoing shipping agreement by clicking  next to the agreement in the Outgoing Shipping Agreements table.

Click  in the batches column to open a table of all the batches created and shipped under that agreement.

You can only make changes while the game is running.

New Shipping Agreements

There are two ways to create a shipping agreement:

- Click **ADD NEW +** at the top-right of the Outgoing Shipping Agreements table, then follow the prompts to create the shipping agreement for that product to that team.
- Click  at the right of an existing shipping agreement in the Outgoing Shipping Agreements table to make a copy of that agreement, then follow the prompts. After the new shipping agreement appears in your Outgoing Shipping Agreements table, you can edit its terms before sending it to the buying team.

The seller proposes the created shipping agreement to the buyer by clicking  next to the agreement in the Outgoing Shipping Agreements table. As soon as the seller clicks , the proposed shipping agreement appears in the Incoming Shipping Agreements table of the buyer's DC. When a DC has received a new proposal or counter proposal, its icon turns yellow.

To review the terms of a proposed shipping agreement, click the Terms box in the Shipping Agreements table. Click  in the Shipping Agreements table to accept,  to reject, or edit the terms and send the agreement back to the other party to make a counter proposal.

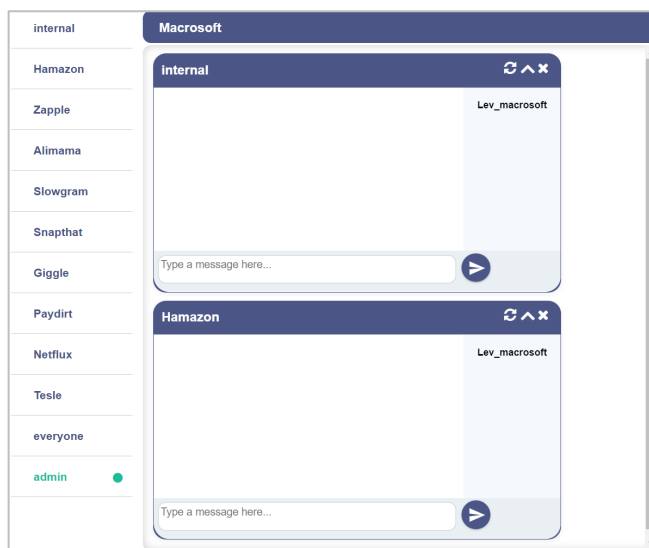
When a shipping agreement is proposed or counter proposed, clicking  next to it in the Shipping Agreements table rejects or rescinds the agreement. When a shipping agreement is active, clicking  terminates it permanently.

Shipping Agreement Terms

- **Order quantity** specifies the number of units in the shipment. If the shipment is triggered but the seller does not have enough units to ship, it waits until enough units are available.
- **Unit price** is the price per unit paid by the buyer to the seller when the inventory is shipped.
- **Ship mode** specifies whether the units are shipped by mail or container. Each batch is shipped on its own. Two or more batches cannot be combined in a single shipment. Mail is faster but usually more expensive per unit. Containers are slower but can be less expensive. Containers hold up to 1,000 units and mail is sent in packages of 10 units. If you send fewer units, you pay for the full container or package. Costs and transit times are as follows:
 - **Container within a region:** free; 1 day
 - **Container between regions:** \$10,000 per 1,000 units; 21 days
 - **Mail within a region:** free; 1 day
 - **Mail between regions:** \$400 per 10 units; 3 days
- **Franchise fees** are flat fees paid by the buyer to the seller. These can be (1) one-time fees paid at the inception of the agreement, (2) ongoing fees paid at the end of each day the agreement is active, or (3) revenue sharing where the buyer pays a set amount and/or a percentage of the retail price back to the seller each time the buyer sells a unit of the inventory to a retail customer.
- **Slotting fees** are flat fees paid by the seller to the buyer. These can be one-time fees paid at inception or ongoing fees paid at the end of each day the agreement is active.
- The agreement can include minimum and maximum **limits on the reorder point** value that the buyer can set. A minimum reorder point, for example, can ensure that the buyer holds a certain amount of inventory. The agreement can also include **limits on the priority** value that the seller can set and **limits on the retail price** that the buyer can set.
- **Termination** specifies a minimum number of days that the agreement will be active from the moment it is activated. If the either team terminates the agreement before the termination period has elapsed, that team pays the specified penalty to other team. There is also an option to let the buyer return the unsold inventory to the seller at a prearranged price after the agreement is terminated. To initiate the return, the buyer should click on the  icon in the batches column of the Incoming Shipping Agreements table. If returns are allowed there will be a  icon in the batches table that opens.
- The **penalty for late shipments** specifies how many days the seller has to ship an order after the buyer requests it. If it takes longer to ship the order, the price for those units decreases by a specified percentage per day for each day that it is late until the order is shipped. If the penalty reaches 100%, the buyer pays nothing for the order when it is finally shipped. The lateness penalty is prorated over fractions of a day. Both the incoming and outgoing shipping agreement tables include a column showing each agreement's delay at that moment.
- Either team can enter other terms that are binding but must be enforced by the teams instead of the software.
- There is a box at the bottom of the panel where students can enter comments for a particular revision of the terms.

Communication

Click the “Chat Panel” button above the map to open the chat panel window. The chat panel is divided into two sections: a left-hand menu and a chat area to the right of the menu. The menu contains links for the various groups with which you can communicate. There is an “internal” link for your own team, a link for each other team, an “everyone” link for all teams, and an “admin” link for instructors. Clicking on a link opens a chat window for that group in the chat area. You can open more than one chat window at a time. A closed or minimized chat lights up in green when there is a new message.



Other Considerations

If your firm runs out of cash it will receive emergency loans to continue operations. Emergency loans are automatically repaid as cash becomes available. Interest on emergency debt is charged at an annual percentage rate (APR) of 40%.

Cash on your balance sheet earns interest at a 3% APR.

Cost of capital is 10% APR.

Expenses for raw materials are payable 15 days after they are incurred, while other expenses are payable after 30 days.

Profits are taxed at a rate of 35%, payable at the end of every quarter.

Reports

The **Financial Statements** report contains the firm's consolidated income statement, balance sheet, and cash flow statement. Use the dropdowns at the top of the report to select the team, statement, and time period: days, weeks, quarters, and years. Only quarterly and annual statements are available for other teams. Click the \$ symbols in the balance sheet or income statement to switch to a % view and vice versa. Clicking the \$/% at the top of the column switches the entire statement. Download a statement by clicking  at the top right of the page.

The **Working Capital Analysis** report contains the firm's current assets, current liabilities, total working capital and current ratio, non-cash working capital and current ratio, and quick ratio. Use the dropdowns at the top of the report to select the team and time period: days, weeks, quarters, and years. Only quarterly and annual information is available for other teams. Click on an item label to see a chart of the values for that item. Download the report by clicking  at the top right of the page.

The **Market Grid** report contains matrices of various metrics for each product market and geographic region in the game. The metrics are cumulative unit sales, cumulative revenues, average retail price, fraction of the market served, and concentration of sales by team. Select a metric using the dropdown at the top of the report. Download the data by clicking  at the top right of the page.

The **Product Analysis** report shows the current price, current on-hand inventory, and cumulative retail sales for each product at each DC. Use the dropdowns above the report to filter by manufacturer, owner, and region.

The **Price Response** report provides data on average retail units sold per day at each price for a given product in a DC, as well as the ability to plot various relationships in a graph above the data table. Switch to a different product by clicking  next to the product name at the top of the page. Download the data by clicking  at the top right of the page.

The **Inventory Status** report shows a snapshot of your supply chain at a given point in time: batches being produced at the factory, batches in transit to the DC, on-hand inventory levels at the DC, stocked out items, and inactive inventory. The data includes product names, locations, quantities, days remaining until new batches are produced and delivered, and estimated days remaining until on-hand inventory runs out.

The **Inventory Activities** report shows an overview several metrics for the products carried in your DC: retail prices, retail and wholesale revenues and units sold, inventory levels, latest inventory movement dates, recent average unit sales, estimated days until on-hand inventory stocks out, and estimated days until the next batch arrives at the DC. Use the dropdowns above the report to filter by product, manufacturer, and region.

The **Inventory Status Grid** report shows one of several metrics for the products at your DC: running shipping agreements, days supply on hand, units on hand, units in process, units inbound, units total, price, delays, priorities, reorder points, contribution, and service rate. Select a metric using the dropdown at the top of the report. Click on a value to see more details.

The **Active Shipping Agreement Sankey** report visualizes product flows between origin and destination for all of your active shipping agreements. Dropdowns above the report filter by product, focus on active vs. operating agreements, and switch between the order quantity, total batches, total units, and total returns metrics.